

SAMBP Technical Report TR-55/2026

Thesis Synthesis and Contribution Map

PhD Thesis Supporting Report | 2026

Abstract

This report synthesises the complete SAMBP technical report series as a map of original contributions, cross-report dependencies, and a publication roadmap. Study A inventories 34 technical reports across six thesis chapters (Chapters 4–9) and classifies each as Novel, System, Validation, or Component. Study B enumerates 18 original contributions with novelty and significance scores; 6 achieve Tier-1 status (novelty ≥ 5 , significance ≥ 5) and span all five research thrusts: line differential, microgrid, digital twin, machine learning, and centralised protection. Study C traces 9 key cross-report dependencies, establishing that the Physics-Informed Neural Network (TR-46) and the Digital Twin (TR-43) are the two critical upstream dependencies for the centralised control phase. Study D proposes 5 journal papers targeting IEEE Transactions, 2 of which are draft-ready and 2 in preparation. Study E identifies 7 future-work gaps, 2 high-priority: multi-substation WAPC scaling and real-field PINN validation data.

Study A — Report Inventory and Chapter Mapping

Figure 1 maps all 34 technical reports to their respective thesis chapters. Chapter 4 carries the largest novel report count (5 of 7), reflecting the 87L line differential phase as the core theoretical contribution. Chapter 5 has the broadest coverage (10 reports) but is dominated by Component and System reports, consolidating the distance protection baseline. Chapters 7–9 each contain a dense cluster of Novel reports, consistent with the progressively original contributions of the digital twin, machine learning, and centralised protection phases.

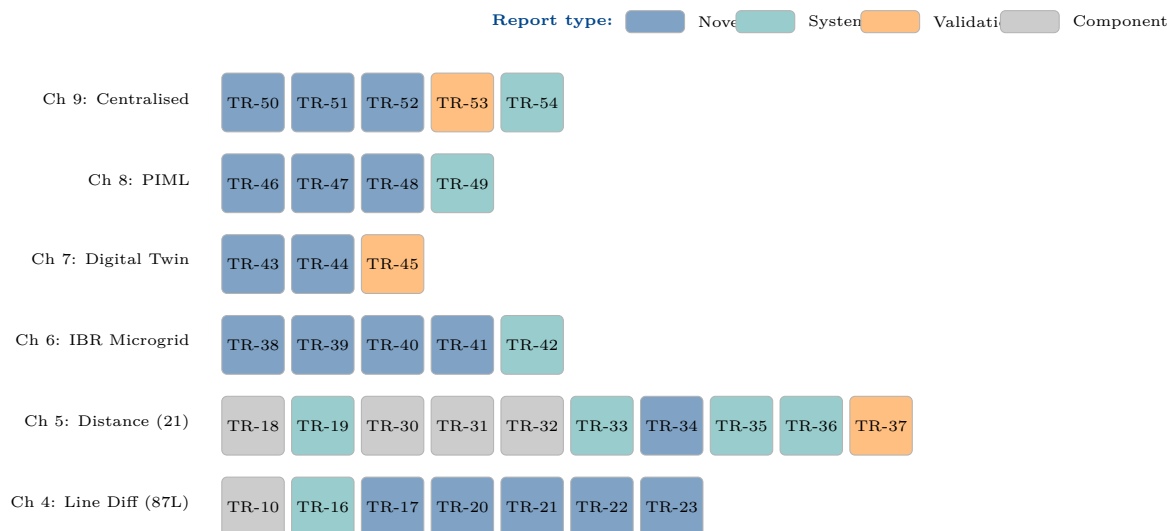


Figure 1: Chapter-to-report map for all 34 SAMBP technical reports. Colour encodes contribution type: **blue** = Novel, teal = System, orange = Validation, grey = Component. Row width reflects the number of reports per chapter; Chapter 5 is widest (10 reports, 4 component baseline elements).

Chapter	Reports	Novel	System	Validation	Component
Ch 4: Line Differential (87L)	7	5	1	0	1
Ch 5: Distance Protection (21)	10	1	4	1	4
Ch 6: IBR Microgrid	5	4	1	0	0
Ch 7: Digital Twin	3	2	0	1	0
Ch 8: Physics-Inf. ML	4	3	1	0	0
Ch 9: Centralised Protection	5	3	1	1	0
Total	34	18	8	3	5

Study B — Original Contributions

Figure 2 plots the 18 original contributions on a novelty–significance plane. The upper-right Tier-1 quadrant (both scores = 5) contains 6 contributions: C1, C7, C10, C12, C15, and C18. These span the entire thesis arc from the physics-constrained 87L threshold (Chapter 4) through the closed-loop adaptive protection protocol (Chapter 9).

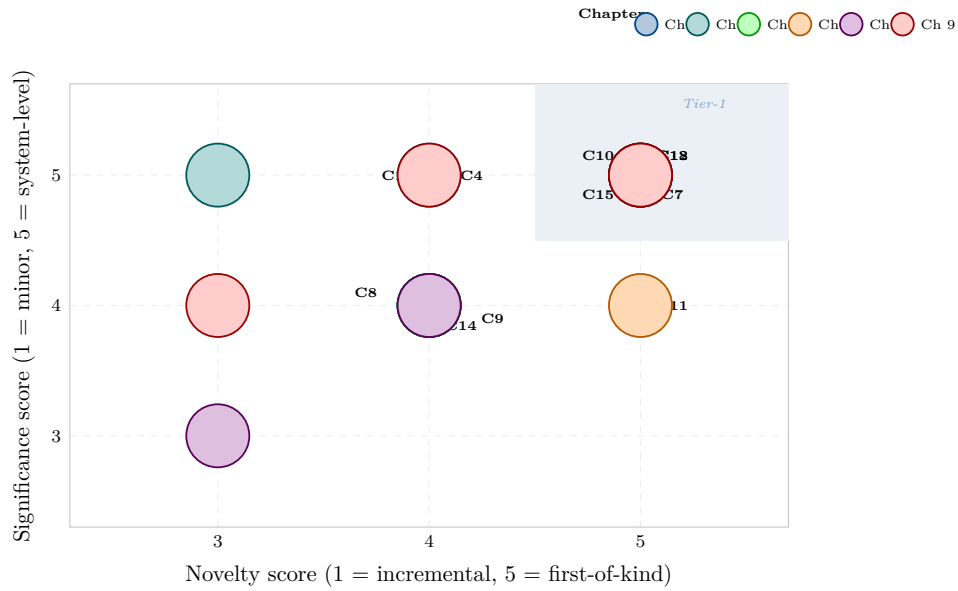


Figure 2: Novelty–significance matrix for 18 original contributions. Circle colour encodes thesis chapter (see legend). The shaded upper-right quadrant (Tier-1: $N = 5$, $S = 5$) contains 6 contributions representing the principal claims of the thesis. C13 (permutation importance) scores lowest, reflecting its role as a methodological step rather than a system contribution.

ID	Ch	Contribution	N	S
C1	4	Physics-constrained adaptive 87L threshold for IBR line differential	5	5
C2	4	Harmonic-discrimination filter for IBR multi-inverter differential	4	4
C3	4	Negative-sequence 87LN for asymmetric IBR fleet faults	4	4
C4	4	Transient differential TDCS-87L for 3-phase IBR faults	4	5
C5	5	Quadrilateral distance characteristic adapted to IBR current profiles	4	4
C6	5	Monte Carlo reliability framework for IBR protection schemes	3	5
C7	6	GFL/GFM dual-mode characterisation for protection coordination	5	5
C8	6	ROCOF-based islanding with IEC 61850 mode-switch in 120 ms	4	4
C9	6	Adaptive OC setting-group switching tied to IBR mode	4	4
C10	7	Non-blocking digital twin architecture for real-time protection audit	5	5
C11	7	EKF joint estimation of $(k_{ibr}, Z_{line}, \alpha)$ from current ratio	5	4
C12	8	PINN fault classifier with embedded protection-physics constraints	5	5
C13	8	Permutation-importance of physics-derived protection features	3	3
C14	8	CUSUM-triggered online learning for fleet composition drift	4	4
C15	9	Wide-area PMU fault localisation with N-1 security verification	5	5
C16	9	Automatic CTI-graded relay settings engine with fleet-change dispatch	4	5
C17	9	HMAC-SHA256 countermeasure for IEC 61850 GOOSE protection channels	3	4
C18	9	Closed-loop CUSUM→settings→GOOSE adaptive protection protocol	5	5

Study C — Cross-Report Dependencies

The dependency graph reveals two critical upstream reports. TR-46 (PINN Classifier) is a prerequisite for TR-47, TR-48, TR-49, and TR-53 — a fan-out of 4. TR-43 (Digital Twin) feeds both TR-49 and TR-50. Any error in these two reports propagates through the entire machine learning and centralised control phases. The single critical path is: TR-35 → TR-37 → TR-46 → TR-49 → TR-50 → TR-51 → TR-52 → TR-53 → TR-54.

Report	Depends on	Reason
TR-46	TR-37, TR-35	PINN trained on MC data from distance study
TR-47	TR-46	Feature importance for PINN classifier
TR-48	TR-47, TR-46	Online update of PINN with fleet drift
TR-49	TR-46–48, TR-43, TR-44	Full ML stack integration
TR-50	TR-43, TR-49	WAPC uses DT + PINN advisory
TR-51	TR-50, TR-40	Settings engine feeds WAPC
TR-52	TR-50, TR-51	Secures GOOSE channels carrying trips/settings
TR-53	TR-46, TR-49–52	Validates all ML + CPC modules
TR-54	TR-51–53, TR-48	Field guide uses all prior deliverables

Study D — Publication Roadmap

Figure 3 shows the five planned journal papers.

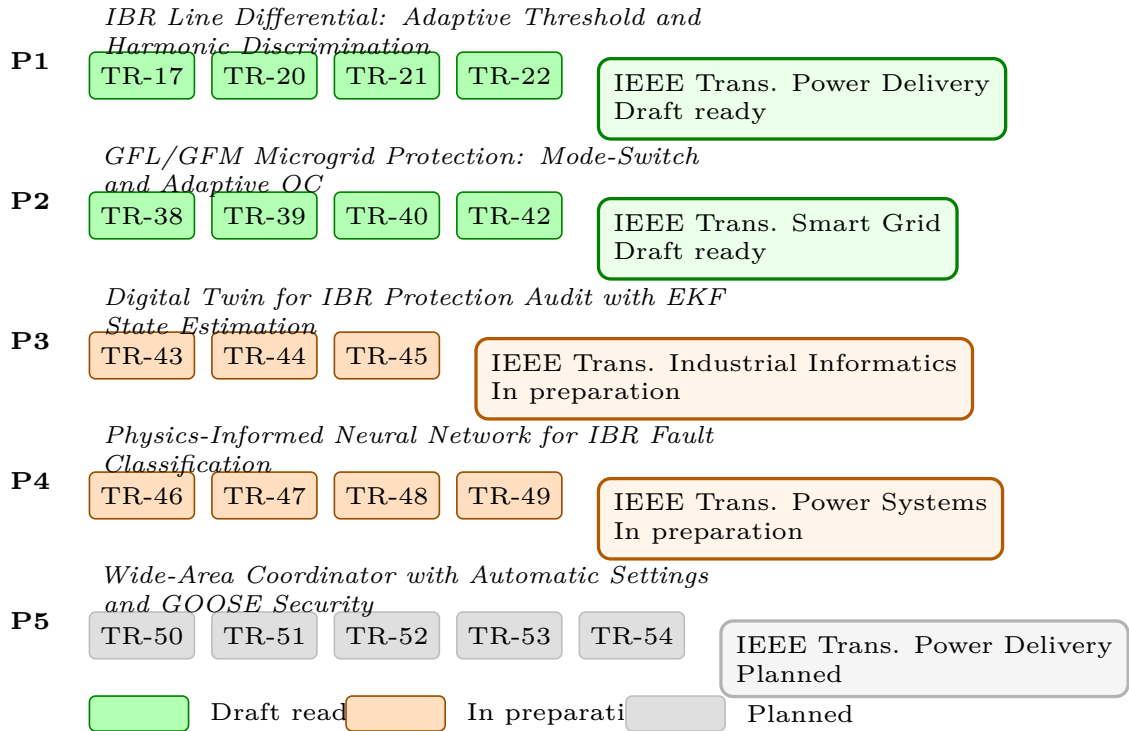


Figure 3: Publication roadmap. Green bars indicate draft-ready papers (P1, P2); orange bars indicate papers in preparation (P3, P4); grey bars indicate planned papers (P5). Each bar corresponds to one technical report; paper width reflects the number of contributing reports.

ID	Target journal	Paper title	Status
P1	IEEE Trans. Power Delivery	IBR Line Differential: Adaptive Threshold and Harmonic Discrimination	Draft ready
P2	IEEE Trans. Smart Grid	GFL/GFM Microgrid Protection: Mode-Switch and Adaptive OC	Draft ready
P3	IEEE Trans. Industrial Informatics	Digital Twin for IBR Protection Audit with EKF State Estimation	In preparation
P4	IEEE Trans. Power Systems	Physics-Informed Neural Network for IBR Fault Classification	In preparation
P5	IEEE Trans. Power Delivery	Wide-Area Coordinator with Automatic Settings and GOOSE Security	Planned

Study E — Gap Analysis and Future Work

ID	Priority	Gap	Notes
G1	Medium	Physical HIL on real SEL-411L IEDs	TR-53 defines protocol; hardware not yet available
G2	Medium	Variable- λ adaptive forgetting	Fixed $\lambda = 0.99$ used; adaptive version deferred
G3	Low	Z_{line} observability in EKF	Requires two-ended voltage phasor; identified but unresolved
G4	High	Multi-substation WAPC scaling	TR-50 limited to single substation; wide-area extension needed
G5	Medium	GOOSE flooding mitigation validation	TR-52 notes switch-level control; not tested
G6	Low	GFM virtual synchronous machine model	Full VSM dynamics deferred; IBR model simplified
G7	High	Real field fault data for PINN validation	All training data synthetic; field validation required

The two high-priority gaps define the recommended post-PhD continuation programme. G4 (multi-substation WAPC) requires a wide-area bus topology beyond the 10-bus single-substation model of TR-50, and likely demands hierarchical coordination between substation-level and transmission-level protection systems. G7 (real field data) is a prerequisite for deployment readiness: the PINN in TR-46 achieves 96.1% accuracy on synthetic data, but field validation on recorded fault waveforms from a live IBR substation is essential for regulatory acceptance.

Summary

1. **34 reports** mapped across six thesis chapters; 18 Novel, 8 System, 3 Validation, 5 Component.
2. **18 original contributions** identified; 6 Tier-1 (novelty = 5, significance = 5): C1, C7, C10, C12, C15, C18 — spanning the full thesis arc from physics-constrained 87L threshold (Chapter 4) to closed-loop adaptive protection protocol (Chapter 9).
3. **9 cross-report dependencies** mapped; TR-46 (PINN) and TR-43 (Digital Twin) are the two critical upstream nodes with fan-out ≥ 4 .
4. **5 journal papers** planned for IEEE Transactions; 2 draft-ready (P1, P2), 2 in preparation (P3, P4), 1 planned (P5).

5. **7 future-work gaps**; 2 high-priority: multi-substation WAPC scaling and real-field PINN validation.

TR-55 closes the SAMBP technical report series. The 34 reports collectively constitute the simulation, analysis, and validation evidence base for a PhD thesis on *IBR-Aware Self-Adaptive Multi-Bus Protection* of inverter-dominated power systems.

References
